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Negotiation as an interpersonal skill: Generalizability of negotiation outcomes and tactics across contexts at the individual and collective levels

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ABSTRACT

As part of a program of research to develop and evaluate assessments of interpersonal skills, we developed a dyadic negotiation task that allows participants to select options and to communicate with each other through a chat window interface. Based on the New Recruit negotiation task, we developed an assessment of negotiation skills that included three negotiation issue types—distributive (zero-sum), integrative (trade-off), and compatible (aligned)—implemented in two tasks (Fundraiser and Robot) given to 84 participants in dyads. We found evidence for the generalizability of negotiation skill across tasks at both the individual ($r = 0.62$) and joint ($r = 0.74$) levels. Participants' chat turns were coded into 14 tactic categories (e.g., making a concession, stating a disagreement). Tactic use across tasks was consistent at both individual (average correlation was $r = 0.52$), and joint ($r = 0.58$) levels. Principal component analysis of tactics used identified four tactic types (cooperative, competitive, greetings, impasse). Greater use of cooperative tactics was associated with higher joint negotiation scores (i.e., sum of individual scores). This research contributes to an understanding of negotiation as a general individual and joint skill and suggests specific tactics associated with that skill. The research also highlights the value of an online platform for conducting such research that enables small-group interactive task performance in which participants communicate with each other using a text-based chat box.

1. Introduction

Interpersonal skills, including negotiation skills, communication, and collaborative problem-solving skills are important skills for school and the workplace. Employer surveys conducted within and outside the U.S. have identified interpersonal skills such as communication and problem solving as among the most highly sought skills in new workers (Casner-Lotto & Barrington, 2006; McKinsey, 2012; NACE, 2015). Interpersonal skills are likely to become increasingly valued and rewarded in the workforce due to changes in technology (Deming, 2017). As technology replaces routine and increasingly non-routine cognitive work (Autor, Levy, & Murnane, 2003; Elliott, 2017; Frey & Osborne, 2013), hard-to-automate skills, such as interpersonal skills, particularly in conjunction with complex cognitive skills such as problem solving, will likely be the ones that remain viable in the future workforce (Deming, 2017; Weinberger, 2014).

An important kind of interpersonal problem-solving skill is negotiation. Negotiate tasks represent a full quadrant of McGrath (1984) group tasks taxonomy (the others being generate tasks [e.g., idea

generation], choose tasks [e.g., achievement tests], and execute tasks [e.g., crew performance]). Negotiation courses are among the most popular courses in business schools, reflecting the importance of negotiation not only in commerce but in all areas of life where just about any situation can be seen as a negotiation (Thompson & Leonardelli, 2004). Negotiation has been studied from the standpoints of not only business, but also social psychology, game theory, and behavioral economics (Bazerman, Curhan, Moore, & Valley, 2000; Thompson, Wang, & Guinia, 2010). Negotiation tasks are an interesting class of interpersonal skills tasks in that they represent a fairly large swath of interpersonal tasks and thus gaining insight into negotiation task behavior is likely to pay dividends in understanding social and collaborative skills more generally. The broader domain of social and collaborative skills has also been studied from a variety of perspectives including educational and organizational psychology (e.g., von Davier, Zhu, & Kyllonen, 2017) and from the small groups' perspective of social psychology (e.g., Larson, 2010).

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1.1. Negotiation definitions, strategies, and tactics

Negotiation can be defined as a process through which two or more parties interact to create potential agreements intended to provide guidance and regulation to their future behavior (Sawyer & Guetzkow, 1965). Negotiation is a “ubiquitous social activity” that occurs “anytime people cannot achieve their goals without the cooperation of others” (Thompson, Wang, & Gunia, 2010, p. 492).

Negotiations usually involve different issues that must be agreed upon. The literature distinguishes three kinds of issues (ten Brinke, Black, Porter, & Carney, 2015; Gelfand, Fulmer, & Severance, 2011). The first is *distributive* issues (also “zero-sum”) in which negotiators view the task as one involving either winning or losing with their goal being to win (Gelfand et al., 2011). The second is *integrative* issues in which the parties can maximize joint gains by searching for an outcome in which both parties benefit. This typically involves a tradeoff in which parties gain ground on issues important to them in exchange for giving ground on issues less important to them (Elfenbein, Curhan, Eisenkraft, Shirako, & Baccaro, 2008). The third is *compatible* issues in which the priorities of the negotiating parties are aligned, and interests are mutual (Thompson & Hrebec, 1996). In this study we were interested in negotiation behavior with respect to all three kinds of issues, and so we purposely set up payoffs to make the negotiations feature distributive, integrative, and compatible issues.

Negotiation can also be understood from the standpoint of game theory (e.g., Hausken, 1997; Turocy & von Stengel, 2002). One useful concept from game theory is that of Pareto optimality, which refers to the allocation of resources (in the tasks we administer here, points) across parties so that no other allocation can make a party better off without hurting another party. For two of the issue types we present (integrative and compatible), achieving Pareto optimality is the unstated goal of the negotiation (for zero-sum or distributive issues, all states are Pareto optimal).

Real-world negotiations often feature opportunities for mutual benefit and opportunities for personal gain at the expense of the other party (Deutsch, 1949; Tjosvold, 1984). Competitive negotiation strategies are those aimed at maximizing individual gains (De Dreu, Weingart, & Kwon, 2000) and include tactics such as creating an impasse (i.e., refusal to reach an agreement), staking out a strong starting position, and disagreeing with an offer proposed by the other party. Cooperative negotiation strategies, those aimed at maximizing joint gains (De Dreu et al., 2000), include tactics such as making a concession to the other party, establishing a compromise, or “logrolling,” in which parties engage in strategic trade-offs that differ in priority for each negotiator such that it is possible to yield on lower priority issues in exchange for maximizing higher priority ones (Pruitt, 1981).

1.2. The space of negotiation and mixed-motive tasks

As a “ubiquitous social activity” the space of negotiation tasks is potentially quite large. From the perspective of McGrath's taxonomy, negotiation tasks are those that involve conflict (vs. cooperation), but such tasks may be difficult to distinguish from decision-making tasks, at least in natural settings (Larson, 2010). Negotiation tasks are also often referred to as mixed-motive tasks in that individuals within the negotiating dyad must resolve conflicts of interest in which the interests of the group (i.e., joint interests) can conflict with individual interests. As such, many economic games are often considered negotiation tasks in experimental studies. These include prisoner's dilemma and variants (Halevy, Chou, & Murnighan, 2012), coalition games (Van Beest, Van Kleef, & Van Dijk, 2008), and ultimatum games (e.g., Van Dijk, van Kleef, Steinel, & van Beest, 2008).

Negotiation games are commonly used as exercises in business school to familiarize students with issues and to teach strategies. Kellogg School of Management (2008) produced a brochure identifying a number of negotiation (Thompson, 2012) and decision making (based

on Murnighan, 1991) exercises that could be used for this purpose. It is useful to review the types of games and exercises offered to get a sense for the space of negotiation tasks. Exercises fall into the categories of decision-making, negotiation (cross-cultural, dispute-resolution, third party, and two-party deal making), and teams exercises (creativity, cross-functional teams, team decision making, team dynamics). The dispute resolution and two-party deal making negotiation tasks are closest in intent to what we are examining in this study. Dispute resolution (12 tasks) includes exercises designed to teach distributive strategies, distinguish rights and interests, understand emotional aspects, and deal with multiple issues. Two-party deal making (35 tasks) exercises address integrative and distributive strategies, ethics, issues of trust and misrepresentation, and introduce concepts such as opening offer, best alternative to a negotiated agreement (BATNA), resistance and aspiration points, and bargaining zone. Exercises in both categories are presented with various cover stories including real estate developer vs. subcontractor, condominium owners, repair bill disputes, roommate rent splitting, buyers and sellers of various goods and services, union negotiations, hiring interview, and many others. One of the negotiation tasks is a multi-trial prisoner's dilemma game set in a corporate context, illustrating the connection between the two domains.

Negotiation tasks include one called New Recruit (Neale, 1997) (also a variation called Outside Offer; Maddux, 1997), which is characterized as “two-party, multi-issue, quantified negotiation over an employment contract. The exercise illustrates Pareto optimality and the differences between compatible, trade-off or integrative, and distributive issues” (Kellogg School of Management, 2008). Perhaps because of its comprehensiveness in touching on these issues and its prototypical nature, New Recruit is frequently used as a learning exercise for business students and as a task in negotiation research (Elfenbein et al., 2008). For this reason, we chose to develop variants of the New Recruit task for this research.

1.3. Individual differences in negotiation skill

Bazerman et al. (2000) concluded in their review that there was little evidence for the importance of individual differences in negotiation skill in that personal characteristics did not account for much variance in negotiation outcomes and context swamped individual differences in importance. An example of a context manipulation is labeling a repeated-play prisoner's dilemma game as a “Wall-street game” versus a “Community game” which led to twice as much cooperation in the Community game (Ross & Ward, 1995). However, Bazerman et al. (2000) did concede the potential importance of cultural differences, such as an individualistic vs. collectivist orientation (Markus & Lin, 1998), and more recent research has explored the importance of personality (Wilson, DeRue, Matta, Howe, & Conlon, 2016), and other personal factors (Elfenbein et al., 2008), such as attitudes toward negotiations, conflict style, intelligence, personality, and demographic characteristics. This also includes dyadic interactions (Elfenbein, Eisendraft, Curhan, DiLalla, 2018) and training effects (Stevens et al., 2018). This more recent research has suggested a much larger role for individual differences; for example Elfenbein et al. (2008) estimated that over 40% of the variance in performance outcomes could be attributed to individual differences.

Context effects and individual differences are not opposed to each other, and both can contribute to task performance, simultaneously (Steyer, Mayer, Geiser, & Cole, 2015). It is difficult to imagine that negotiation is not a skill, and that some individuals are not better at it than others. Although individuals may differ in what they hope to achieve as a result of a negotiation, in all negotiations the goal is to achieve an optimal solution that maximizes value for both parties (ten Brinke et al., 2015). Thus, if value for each party is quantified, individual and joint gains can be calculated and used as measures of negotiation skill at the individual and joint levels.

In classical test theory, reliability, or more generally,

generalizability (Webb, Shavelson, & Haertel, 2006) is a direct measure of individual differences in a skill. The generalizability question is what proportion of variation in observed task outcomes is associated with individual differences (individual differences are reflected in true scores, or universe scores) as opposed to task or context differences (which are reflected in that portion of variance in observed scores not associated with true scores, sometimes referred to as error or measurement error), and what are the conditions or contexts over which the generalization occurs. For example, it would be possible in principle to generalize across all negotiation tasks; in this study we confine generalization to two negotiation tasks that have similar formal structure of issues and payoffs, but that vary in their settings. Generalizability can be quantified with a correlation across tasks, or subtasks, at the individual and joint levels. Individual gains represent the value of the negotiated agreement to one individual negotiator while joint gains refer to the sum of the gains obtained by both negotiators.

1.4. Negotiation findings

Key findings from negotiation research have been summarized in several review articles (Bazerman et al., 2000; Thompson et al., 2010). Main effect findings most pertinent to the negotiation tasks here are that negotiators often assume a fixed pie and “miss opportunities for mutually beneficial trade-offs between parties” (Bazerman et al., 2000, p. 283). Negotiators often falsely assume that their preferences for the negotiation are opposed to the preferences of the other party, and they often ignore the perspective of other parties, for example by devaluing concessions made by the other (Bazerman et al., 2000). Negotiators also often tend to “leave value on the table,” failing to reach mutually beneficial agreements due to incorrect assumptions about the negotiating partner (Thompson & Lucas, 2014). These factors suggest that parties will often fail to find common ground on compatible issues and settle for non-optimal solutions, for example, failing to find an integrative solution when possible (Thompson et al., 2010).

1.5. Study purpose

The purpose of this study was twofold. First, we sought to evaluate the generalizability of negotiation skills across two negotiation tasks at both the individual and joint level. Negotiation tasks are mixed-motive tasks in that participants are motivated to achieve both personal gains (i.e., accumulate more points personally) and a successful joint outcome (i.e., accumulate more total points as a dyad). For some issues within our negotiation tasks the individual and joint motivations are aligned, and for some they are in conflict. The question of generalizability concerns whether individuals who perform well on one task (i.e., achieve individual gains, accumulate more points) perform well on the other; and whether dyads that perform well on one task (i.e., experience successful joint outcomes; accumulate more points as a dyad) perform well on the other. It is possible that there can be generalizability (i.e., high correlations across tasks) at one unit of analysis (e.g., individual) but not at the other (e.g., joint) or that there can be generalizability at both (i.e., high correlations between individual outcomes across tasks, and joint outcomes across tasks), or neither. To our knowledge, this specific issue of generalizability of negotiation skill at the individual and joint levels has not yet been evaluated. The question of generalizability has been addressed for other small group tasks (e.g., Woolley, Chabris, Pentland, Hashmi, Malone et al., 2010), where there has been evidence for generalizability at the joint level. However, mixed-motive tasks are likely different because there can be a conflict in incentives at the individual and joint level.

The second purpose of the study was to explore process consistency across tasks, and process correlates of negotiation success, separately at the individual and joint levels, and separately by issue type. We operationalized process by frequency of specific tactics used in negotiation discussions. Our negotiation task was administered online, and

participants communicated through a chat window. Each chat turn was classified with respect to 14 tactic categories (e.g., create an impasse, articulate a starting position). We evaluated process consistency by correlating tactic use across tasks. We evaluated process correlates of negotiation success by correlating tactic use with points at both the individual and joint levels. These approaches allow us to address questions such as, are specific negotiation tactics, such as refusing to reach an agreement, associated with personal gains in the negotiation? Are others, such as logrolling (trading off issues), associated with joint gains, particularly for integrative and compatible issues? Among the tactic categories were maintaining communication tactics, such as greetings and affect displays, which are thought to generate and maintain social bonds (Nardi, 2005), and which we have identified in prior collaborative research (Hao, Liu, von Davier, & Kyllonen, 2017). There have been process analyses of small group task performance, for example, finding turn-taking to be associated with joint success (e.g., Wooley et al., 2010). However, to our knowledge, an individual and joint generalizability analysis based on negotiation tactics has not been conducted for negotiation tasks.

2. Method

2.1. Sample

The sample was 42 dyads, comprised of 84 undergraduate students participating online (through Amazon Mechanical Turk). Participants were recruited from an online panel and paid \$9–\$11 US per hour for participating. Each session took a little less than an hour.

The sample identified as 51% female; 59% White, 22% Asian or Asian American, 13% Black or African American, 13% Hispanic or Latino, 8% American Indian or Alaska Native, and 3% as Native Hawaiian or Pacific Islander (participants could select more than one race/ethnicity category). A plurality of participants (41%) reported having a college GPA in the 3.5–4.00 range, 31% from 3.00 to 3.49, 15% from 2.50 to 2.99, and 6% from 2.00 to 2.49; 5% were either entering freshmen or did not respond. Mean SAT scores were 698 for Reading, 634 for Mathematics, and 619 for Writing. Mean ACT scores were 22 for English, 21 for Mathematics, 23 for Reading, and 21 for Science.

2.2. Procedure

All study materials were approved by an Institutional Review Board prior to the launch of the study. After agreeing to participate in the study, online participants were administered a background information questionnaire to screen for being current undergraduate students. They then were matched with a dyad partner who chose to participate at approximately the same time. The dyad remained intact over two online negotiation tasks (Fundraiser and Robot). The order of the two tasks was randomized. Participants were paid \$9 US for completing both negotiation tasks, with an additional \$1 bonus payment for each negotiation task won. Thus, each participant could earn a total of \$11 US. Each negotiation task had a 30-min time limit. Average completion time for each negotiation task was 10.3 min. Participants also completed a background questionnaire. Altogether the study took less than 1 h to complete.

2.3. Negotiation tasks (Fundraiser and Robot)

We created two negotiation tasks for this study: a Fundraiser task (see Appendix A) and a Robot task (see Appendix B). The general structure of the negotiation tasks was similar to that of the New Recruit exercise (Neale, 1997), used formatively in business schools. In New Recruit there is a clearly defined point structure and three issue types: integrative (“trade-off”), distributive (“zero sum”), and compatible; that structure was followed here.

Fig. 1. Screenshot of the Robot negotiation task.

In our version, the Fundraiser task required students to imagine they were organizing a fundraiser for charity. Issues up for negotiation were (a) the type of fundraising event, (b) the division of responsibilities, (c) when to meet to plan the fundraiser, (d) the charity that would be the beneficiary, (e) when in the semester to hold the fundraiser, and (f) the fundraising target amount. The Robot task required students to imagine they were planning to enter a robot design competition as a team. Issues up for negotiation were (a) choosing the robot's name, (b) the division of responsibilities, (c) selecting the robot's size, (d) choosing the competition to enter, (e) determining who would handle the robot's design, and (f) determining who would handle the robot's parts. A screenshot of one of the tasks is presented in Fig. 1.

Although tasks differed in the surface level goal of the negotiation and the particular issues, they contained the same issue structure and similar payoff schedules. Individuals who earn more points in the negotiation tasks obtained agreements that were more favorable for themselves personally, and as a result, were able to earn more pay for their performance in the negotiation. Thus, points earned in the

negotiation are a measure of a participant's negotiation skill, with more points indicating greater negotiation skill. Each task contained six issues for negotiation: two compatible, two integrative, and two distributive issues. Each issue had five possible options that could be agreed upon. Participants were presented with a point structure that was visible only to them. The point structure indicated how many points each of the issues and associated options were worth. The highest number of points that could be obtained from the Fundraiser task was 9600 and the lowest was - 8400; for the robot task the maximum number of points was 9200 and the lowest was - 8400. Participants were warned not to share any information from their point structure with their partner at any time during the tasks.

To maximize joint outcomes, each partner should give up on the integrative issues less important to them and gain on the issues more important. Trading off in this way, along with selecting the highest point options for the compatible issues, and splitting the difference on the distributive issues would result in a joint outcome of 6800 to 7600 points. Merely splitting the difference on each issue would result in a

Table 1

Payoff schedule for each partner (P1, P2) by issue type (Distributive, Compatible, Integrative).

Distributive (Zero-Sum) Issue			Compatible Issue		Integrative Issue of Value to P1		Integrative Issue of Value to P2	
	P1	P2	P1	P2	P1	P2	P1	P2
Option 1	0	2400	Option 1	0	Option 1	4000	Option 1	0
Option 2	600	1800	Option 2	400	Option 2	3000	Option 2	400
Option 3	1200	1200	Option 3	800	Option 3	2000	Option 3	800
Option 4	1800	600	Option 4	1200	Option 4	1000	Option 4	1200
Option 5	2400	0	Option 5	1600	Option 5	0	Option 5	1600

Table 2
Coding rubric for categorizing negotiation tactics.

Negotiation Tactics	Definition	Actual chat examples from both tasks	Inter-rater Reliability (ICC2)
Cooperative Tactics			
1. Concession	Give in to partner's suggestion	- "I'll go with whatever you are fine with." - "I don't mind 'Eraser.'"	0.83
2. Compromise	Suggest/agree to a compromise	- "So we're splitting down the middle?" - "Well, I really prefer 'Eraser' but I'm trying to meet halfway."	0.81
3. Information Sharing	Provide information about my priorities	- "No, evenings are bad. I'm good with afternoons or lunch time." - "Hmm, 'Destroyer' is a good name."	0.81
4. Stating Agreement	Explicitly state agreement	- "Yeah, I'll go with that!" - "Ok, I agree on those two issues."	0.85
5. Feelings/thoughts Inquiry	Ask partner about thoughts/feelings	- "What are your feelings on that?" - "What about the division of responsibilities? Do you mind taking care of more?"	0.44
6. Trading/log-rolling	Suggest trade-off – if I have first choice on one issue you have first choice on another	- "I will concede on that one if you go with 'A' on the next one." - "To make it easier and faster for both of us I would suggest that you can choose the first three issues and I choose the last three."	1.00
Competitive Tactics			
7. Impasse	Refuse to reach agreement	- "You just need to submit. We clearly are not going to compromise." - "I totally disagree."	1.00
8. Seeking Agreement	Explicitly seek partner's agreement	- "The 'mantisweight' seems to be best for me. Same for you?" - "For the robot name ... 'Destroyer' is ok?"	0.86
9. Stating Disagreement	Explicitly state disagreement	- "Sorry, but that seems too lame." - "I think \$500 is a little too low. Students may not take it as seriously. I'm between \$1000 and \$2500."	0.84
10. Starting Position	Make initial suggestion for an issue	- "I think that for me the best option is 'Eraser.'" - "For the fundraising goal, I think either \$500 or \$1000 is good."	0.94
11. Elaboration on Position	Follow statement with exploration/elaboration	- "I think \$2500 is too steep for a raffle." - "Labor Day is a whole year away! I'd really like to raise money for victims of hurricanes soon. They need it now, not next year."	0.57
Maintaining Communication			
12. Greeting/small talk	Greet partner or engage in small talk	- "Hey" - "Awesome. Nice working with you."	0.87
13. Initial Coordination	Coordinate issues to be discussed	- "Okay, so I think the easiest thing to start with is the name. Sound good?" - "Do you want to start with the name?"	0.89
14. Affect Displays	Express feelings verbally or with emojis	- "Cheers! ☺" - "That was easy, LOL."	0.88

joint outcome of 400–800 points, depending on the task. The payoff schedule for the negotiation tasks is displayed in Table 1.

The negotiation tasks were deployed and administered through the ETS Platform for Collaborative Assessment and Learning (EPCAL, Hao et al., 2017b). In the online platform (see Fig. 1), participants could negotiate with one another using a text-based chat box. It was also possible to make and revise option selections for each issue throughout the negotiation. Option selections could be seen by both participants in a dyad. For the negotiation task to be completed successfully, both participants had to select and agree on the exact same options for each issue; thus, it was not possible for participants to complete the task without reaching agreement on all issues (i.e., there was no possibility of walking away from a deal).

2.4. Annotation

In addition to scoring negotiation performance according to the individual and joint points earned, we also annotated the text-based chat communication between negotiators during the tasks. To do this, we developed a coding rubric identifying negotiation tactics participants could employ throughout the negotiation (see Table 2). The unit of analysis was each turn of a conversation or text-based utterance. Four researchers with training in psychology were trained on how to use the rubric. They participated in two rating calibration sessions to ensure coding consistency. Two of these raters double-coded a subset of data from 20% of the dyads so that inter-rater agreement could be calculated. The remainder of the data were individually coded. Using the rubric we developed, it was possible for each utterance to be categorized into multiple of the 14 negotiation tactics in the rubric. Inter-

rater reliability estimates calculated as intraclass correlations (ICC2) are displayed in Table 2. The vast majority of categories yielded what can be considered excellent inter-rater reliability (values ranging from 0.75 to 1.00; Cicchetti, 1994). Only two categories fell below the 0.75 threshold: Elaboration on position produced good agreement (0.57) while feelings/thoughts inquiry yielded fair agreement (0.44; Cicchetti, 1994). In the later analyses, we use the total number of occurrences of each of the coded tactics to characterize the individual and team negotiation process.

3. Results

The results are organized as follows. First, we examine the generalizability of individual and joint negotiation skill operationalized as individual and joint points earned across tasks. Following this, we examine the generalizability of negotiation tactics operationalized as annotated categorizations of chat-based communication during the negotiation, at the individual and joint levels. We then report the results of a principal components analysis of tactic frequency for the 14 tactics at the individual level in which we identify four components. We then report the results of correlating the four component scores with points earned at the individual and joint levels. Finally, we examined the association between tactic use and points earned for different issue types within each task.

3.1. Generalizability of negotiation skill across tasks

3.1.1. Individual level

Mean differences and correlations between scores (points earned)

Table 3
Descriptive Statistics, t-tests, and Correlations between Task Type and Task Order.

Type of Gains	Task	N	Mean	SD	t-statistic	p (t-test)	Pearson correlation	p (correlation)
Individual	Fundraiser	80	1728.05	2975.84	−1.25	.214	.62	.000†
	Robot	80	1529.27	3102.87				
	First	80	1605.95	2979.54	−.93	.357	.62	.000†
	Second	80	1652.50	3105.41				
Joint	Fundraiser	42	3295.35	2762.93	−2.36	.023	.74	.000†
	Robot	42	2916.28	3344.50				
	First	42	3065.91	3520.50	−1.71	.095	.72	.000†
	Second	42	3147.62	2520.28				
Individual	Fundraiser First	38	1745.00	2960.68	−.19	.852	.48	.002
	Robot Second	38	1586.84	3230.36				
	Robot First	42	1479.55	3025.13	−2.07	.045	.79	.000†
	Fundraiser Second	42	1711.91	3026.00				
Joint	Fundraiser First	20	3323.81	2574.08	−.38	.712	.55	.012
	Robot Second	20	3015.00	1942.40				
	Robot First	22	2830.44	4252.00	−3.74	.001	.88	.000†
	Fundraiser Second	22	3268.18	2992.59				

Note. † denotes significance when using the false discovery rate (FDR) control procedure, using a threshold value of 0.05, designed for use in cases of multiple comparisons (Benjamini & Yekutieli, 2001).

for the different task types and task orders are presented in Table 3. We examined the relationship between performance on the Fundraiser and Robot tasks at the individual level finding a correlation of $r = .62$ ($p = .000$) between individual points accumulated across the two tasks. This is evidence for a general, individual negotiation skill—individuals accumulating more points on one task tended to accumulate more points on the other task. We also found a correlation of $r = 0.62$ ($p = .000$) between tasks given in the first versus second position (regardless of task), another approach for assessing generalizability. We tested for differences between task type and type order using paired samples t-tests. We found no difference in scores on the Fundraiser and Robot tasks ($t(79) = -1.25, p = .21$), and no difference in scores for the task presented second versus first ($t(79) = -0.93, p = .36$). Because of the interdependency of the two participants on each other for their performance, these results cannot be interpreted as failure to improve individual negotiation skill.

3.1.2. Joint level

The correlation between the joint mean scores on the two tasks was $r = 0.74$ ($p = .000$), suggesting a joint negotiation skill—dyads performing jointly well (i.e., accumulating more total points) on one task tended to perform jointly well on the other (see Table 3). A similar correlation between joint scores for the first versus second task was found ($r = 0.72, p = .000$), also supporting generalizability of joint negotiation skill. At the joint level, scores were higher on the Fundraiser task compared to the Robot task ($t(41) = -2.36, p = .023$) but not significantly higher on the task that was presented second versus the one presented first ($t(41) = -1.71, p = .095$).

3.2. Generalizability of negotiation tactics used

3.2.1. Individual level

We computed correlations to index the consistency in negotiation tactic frequency across the two tasks, at both the individual and joint levels. At the individual level, the mean correlation between tactic use on one task and tactic use on the other was $r = 0.52$ ($p = .000$) across the 14 negotiation tactics. This indicates consistency in tactic use across tasks at the individual level and serves as additional evidence for the generalizability of negotiation skill at the individual level. Three of the 14 tactics were not significantly correlated across tasks: they were *impasse* ($r = -.02$), which occurred only six times total across the 84 participants, *initial coordination* ($r = 0.27, p = .08$), and *greeting/small talk* ($r = 0.16, p = .29$). These latter two likely would occur at the beginning of the first task and not necessarily be repeated at the second task when participants are already acquainted.

3.2.2. Joint level

At the joint level, mean correlation between tactic use on one task and tactic use on the other was $r = 0.58$ ($p = .000$). (See note above on the low base rate frequency of *impasse*.) This indicates generalizability of tactic use at the joint as well as individual level. There were statistically insignificant correlations for *initial coordination* ($r = 0.27, p = .08$) and *greeting* ($r = 0.16, p = .29$), again, probably due to the lack of need to repeat greetings for the two tasks.

3.3. Dimensions of negotiation tactics

We conducted a principal component analysis of the 14 negotiation tactics, pooling across the two tasks. Based on an examination of the scree plot and the eigen-value-greater-than-one rule (Kaiser, 1960), we extracted four factors. Varimax rotation of the matrix for orthogonal transformation was performed, and rotation converged in five iterations. The results of this analysis are displayed in Table 4. The four factors can be interpreted tentatively as follows: Factor 1: Cooperative tactics; Factor 2: Competitive tactics; Factor 3: Felicitatation tactics; and Factor 4: Impasse tactics. Several tactics have cross loadings. For example, *Information Sharing* and *Compromise* have loadings on both Factors 1 and 2, suggesting that these tactics might be considered both competitive and cooperative tactics.

3.4. Tactic component score correlations with negotiation outcomes

We correlated the four principal component factor scores for the negotiation tactics with individual gains (points) across both tasks at the individual level of analysis (see Table 5). Findings suggest none of the factors are significantly correlated with success in the negotiation at the individual level.

We also correlated the four principal component factor scores for the negotiation tactics with gains (points) across both tasks at the joint level of analysis (see Table 5). Factor 1 (cooperative) negotiation tactics were significantly correlated with greater joint gains ($r = 0.40, p = .000$). Conversely, Factor 3 (felicitatation) negotiation tactics were negatively related to joint gains ($r = 0.24, p = .03$).

3.5. Exploratory analysis of tactic correlations with negotiation outcomes

The component score correlations with negotiation outcomes suggested no tactic categories were associated with success at the individual level, but that cooperative tactics were associated with joint gains. However, we additionally conducted an exploratory analysis with all tactics, not just factor scores, separately for both tasks, and

Table 4
Rotated principal components loadings matrix for negotiation tactic.

Tactic	Factor 1	Factor 2	Factor 3	Factor 4	Chat examples
Stating Agreement	.87				"Yeah, I'll go with that!"
Concession	.85				"I'll go with whatever you are fine with"
Starting Position	.78	.31			"I think that for me the best option is 'Eraser.'"
Seeking Agreement	.72				"The 'mantisweight' seems to be best for me. Same for you?"
Feelings/thoughts Inquiry	.66	.46			"What are your feelings on that?"
Elaborating on Position		.81			"I think \$2500 is too steep for a raffle."
Stating Disagreement		.81			"Sorry, but that seems too lame."
Affect Displays		.77	.36		"Cheers! ☺"
Information Sharing	.57	.67			"No, evenings are bad. I'm good with afternoons or lunch time."
Compromise	.51	.59		.36	"So we're splitting down the middle?"
Trading/Log-rolling		.55		.36	"I will concede on that one if you go with 'A' on the next one."
Greeting/small talk			.79		"Hey"
Initial Coordination	.40		.69		"Okay, so I think the easiest thing to start with is the name. Sound good?"
Impasse				.87	"You just need to submit. We clearly are not going to compromise."

Note. Loadings with values below 0.3 are not listed. Rotated using Varimax rotation with Kaiser normalization.

Table 5
Correlations between negotiation tactic component scores and overall individual and joint gains.

	Individual Gains	Joint Gains
Factor 1	.15	.40**
Factor 2	.09	.15
Factor 3	-.09	-.24*
Factor 4	.09	.13

Note. * $p < .05$, two-tailed; ** $p < .01$, two-tailed.

separately by issue types and at the overall level for each task. As this was an exploratory analysis with many correlations () we used the False Detection Rate (FDR; Benjamini & Hochberg, 1995) approach for establishing the significance of correlations, specifically, a procedure designed for use in cases of multiple dependent tests (Benjamini & Yekutieli, 2001), using an FDR of 0.05. At the individual level the only tactic significantly related to negotiation success was *compromising*, which was negatively related to individual gains on compatible issues in the robot task ($r = -0.45$, $p = .001$). At the joint level the only tactics significantly related to joint negotiation success were *compromising*, which was negatively related to joint gains on compatible issues in the robot task ($r = -0.45$, $p = .001$), and creating an *impasse*, which was associated with joint gains overall across all the issues in the robot task ($r = 0.51$, $p = .000$).

4. Discussion

We investigated negotiation as a general skill at both the individual and joint level. Although there was some evidence for context effects (participants did better as a joint with the more familiar fundraising context), there was strong evidence for consistency or generalizability across tasks at both the individual and joint levels. This implicates both general negotiation skills at the individual level, and general joint negotiation skills at the dyad level. That is, with respect to the negotiation tasks examined in this study, there was evidence for the concepts of a good negotiator and good negotiating teams. This finding, obtained with the present mixed-motive tasks, contradicts the conclusion of no individual differences in negotiation, or individual differences being swamped by contextual effects (Bazerman et al., 2000), and aligns with the findings of individual differences in negotiation (Elfenbein et al., 2008), and with the conclusion of a general collective intelligence factor by Woolley, Chabris, Pentland, Hashmi, and Malone (2010) based on a set of collaborative tasks (e.g., brainstorming, matrix reasoning, planning a shopping trip).

We also found evidence supporting the idea of consistent negotiation tactic use at both the individual and joint levels. The average

correlation of specific tactic use across the two tasks was high at both the individual and joint levels, particularly for certain tactics such as *elaborating on position* at the individual level, and *seeking agreement* at the joint level. Some tactics were used inconsistently, but those could be attributed to low base-rate usage (*creating an impasse* was used only 6 times in total), or the one-time nature of tactics such as *initial coordination* and *greeting* which did not necessarily have to be repeated in a second task due to dyad participants being the same across tasks.

A principal components analysis suggested four distinct factors associated with tactic use. We tentatively defined them as cooperative tactics (Factor 1), competitive tactics (Factor 2), maintaining communication (Factor 3), and impasses (Factor 4). Cooperative tactics include making a *concession*, *stating agreement*, *others' thoughts/feelings inquiry*, *seeking agreement* and establishing a *starting position*. Competitive tactics include *stating disagreement*, and *elaborating* on one's position, and suggesting a *compromise*. The third factor can be characterized by two tactics that maintain communication: engaging in a *greeting* or *small talk* and *initial coordination*. Finally, creating an *impasse* essentially comprises a fourth distinct factor. Given that the sample size for this study was relatively small and that the base rate for some negotiation tactics was somewhat low, future research should further examine the factor structure of negotiation skills.

Finally, we also examined evidence for associations between tactic use and successful negotiation. The main finding here was that although no tactic categories were associated with success at the individual level, cooperative tactic use was associated with success at the joint level. In a more exploratory analysis we found several additional relationships associated with particular issue types for particular tasks that could be explored in future research.

4.1. Limitations

This was a preliminary study and sample size (42 dyads) was small. This resulted in a lack of statistical power to detect true relationships that might have existed between tactic use and outcomes. Future research should replicate our findings using larger samples.

Another limitation to the current study is the use of two fairly similar negotiation tasks. We found that negotiation skill and tactic use generalized across the two tasks, which varied widely in their context. However, the underlying structure of the two tasks was similar and so the conclusion of generalization of negotiation skill is confined to a fairly narrow class of negotiation tasks. Future studies may explore negotiation skills in other contexts.

Additionally, all results pertaining to negotiation performance at the individual level of analysis reflects a dependency upon each participant's negotiation partner within each dyad. Because of the necessarily dyadic nature of a negotiation and the inherent dependency upon the

negotiation partner's performance, it is difficult to disentangle participants' performances from those of their partner. Doing so may require re-pairing individuals with other teammates in dyads. Future research may examine individual contributions this way, and also evaluate the extent to which partner characteristics, such as gender or age, interact to affect individual and group performance.

4.2. Directions for future research

The question of generalization and transfer of negotiation skills is an important one. As noted, the space of negotiation tasks used in this study was narrow; it would be useful to evaluate the generalization of negotiation skills across a much broader range of negotiation tasks. In the introduction we outlined the wide space of negotiation tasks, which also includes mixed-motive decision-making tasks, and opportunities to leave the negotiation for a best alternative to a negotiated agreement, which we did not allow here. A useful and ambitious study could administer a broad set of negotiation tasks (as in the Kellog (2008) exercises) and mixed-motive decision making tasks (e.g., prisoner's dilemma, ultimatum game, coalition game) and explore the factor structure of negotiation skills across those tasks and domains. Studies have explored some of these task domains (Almlund, Duckworth, Heckman, & Kautz, 2011; Falk, Becker, Dohman, Huffman, & Sunde, 2016), but more could be done to map the terrain.

Considerable recent research on negotiation has emphasized personal dimensions to negotiation, such as emotions, race, and gender (Thompson et al., 2010). These are considered most salient when negotiating partners can observe each other, typically face-to-face, at least by video. Here we examined dyadic negotiations conducted using text-based communication only, reducing the potential effect additional information that a visual display would convey (e.g., gender, race, age) along with all vocal and nonverbal cues that often affect communication that occurs in real-life contexts (Knapp, Hall, & Horgan, 2014). Exploring video-based negotiation in future research would allow for

the examination of the effect of vocal and nonverbal cues on negotiation strategies and outcomes. The EPCAL platform we used for this study contains features that would allow for such multimodal communication. EPCAL also enables routine analysis of turn-level timing data, which could supplement the identification of tactics per se as a source of process information about negotiation strategies.

Given that we produced evidence for the generalizability of negotiation skills, a reasonable next step is to explore correlates of skills such as personality, cognitive ability, and other individual difference measures that have been linked to negotiation skills (Elfenbein et al., 2008). We can accomplish that through an expansion of negotiation tasks, per se, but it is also useful to explore correlations with other common-motive small group tasks, such as idea generation, problem solving, and non-mixed-motive decision making. Moreover, future studies may examine the extent to which more experienced negotiators vary from novice negotiators regarding the use of different negotiation strategies.

Finally, in this investigation we relied on human raters to annotate the text-based communication that occurred within the negotiations. This is a time-consuming process, and human biases and errors possibly affect annotations. Given recent advances in engineering and natural language processing, the automated scoring of text-based communication may be a fruitful avenue for future research that may streamline the annotation and assessment process. Automated annotation of the communications (Hao, Chen, Flor, Liu, & von Davier, 2017) may also enable negotiators to receive feedback in or near real-time that could provide opportunities for training and improving one's negotiation skills.

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Appendix A

Fundraiser task instructions for partner 2

Scenario

You and your partner are college students who are organizing a fundraiser for charity. During this task, you will negotiate with your partner about your plans for the fundraiser. You and your partner will need to reach an agreement on the following six issues:

- Type of fundraising event
- Division of responsibilities
- When to meet to plan the fundraiser
- Charity that will benefit from the fundraiser
- When in the semester to hold the fundraiser
- Fundraising target amount

Goal

Your goal is to reach an agreement with your partner that is best for you on all six issues. Refer to the Point Structure, below, to determine how many points each of the issues is worth to you. THE MORE POINTS YOU EARN, THE BETTER.

Point Structure and Fundraiser Task Instructions Charts

Please become familiar with your Point Structure by reviewing the first chart below. Please also review the second chart, the Fundraiser Task Instructions.

Your Point Structure lists the total number of points each issue is "worth." The highest number of points you can obtain from this negotiation (i.e., from all six issues) is 9200 and the lowest number is *negative* 8400. Note that some issues award negative points, which may result in a negative point total. These point totals were calculated by adding up the highest (or lowest) number of points you could receive for each of the six issues.

Important note: Do not, at any time, tell your partner how many points you can receive or share **any** information from your Point Structure. This information is strictly for you.

You will have 30 min to reach an agreement with your partner on all six issues. BOTH YOU AND YOUR PARTNER MUST SELECT MATCHING OPTIONS ON EACH ISSUE FOR THE AGREEMENT TO BE BINDING AND FOR POINTS TO BE AWARDED TO YOU.

There are five different options for each of the issues; these are listed in the Fundraiser task Instructions. Please note the number of points you would receive for each option you and your partner agree on. You and your partner may settle upon any of the five options for each of the issues.

Thus, there is a large number of possible settlements.

Again, remember that your goal is to earn as many points as possible, but that you **must** agree with your partner on each issue in order to earn points.

POINT STRUCTURE		
	Lowest	Highest
Type of fundraising event	0	2400
Division of responsibilities	−6000	0
When to meet to plan event	−2400	0
Charity that will benefit from the fundraiser	0	1200
When in the semester to hold the fundraiser	0	1600
Fundraising goal	0	4000
Total	−8400	9200

Appendix B

Robot Task Instructions for Partner 1

Scenario

You and your partner are college students who want to enter a robot design competition as a team. During this task, you will negotiate with your partner about your plans for building the robot and entering the competition. You and your partner will need to reach an agreement on the following six issues:

- Robot's name
- Division of work responsibilities
- Robot's size
- Which competition to enter
- Robot's design
- Determination of robot parts

Goal

Your goal is to reach an agreement with your partner that is best for you on all six issues. Refer to the Point Structure, below, to determine how many points each of the issues is worth to you. **THE MORE POINTS YOU EARN, THE BETTER.**

Point Structure and Fundraiser Task Instructions Charts

Please become familiar with your Point Structure by reviewing the first chart below. Please also review the second chart, the Fundraiser Task Instructions.

Your Point Structure lists the total number of points each issue is “worth.” The highest number of points you can obtain from this negotiation (i.e., from all six issues) is 9200 and the lowest number is *negative* 8400. Note that some issues award negative points, which may result in a negative point total. These point totals were calculated by adding up the highest (or lowest) number of points you could receive for each of the six issues.

Important note: Do not, at any time, tell your partner how many points you can receive or share **any** information from your Point Structure. This information is strictly for you.

You will have 30 min to reach an agreement with your partner on all six issues. **BOTH YOU AND YOUR PARTNER MUST SELECT MATCHING OPTIONS ON EACH ISSUE FOR THE AGREEMENT TO BE BINDING AND FOR POINTS TO BE AWARDED TO YOU.**

There are five different options for each of the issues; these are listed in the Fundraiser Task Instructions. Please note the number of points you would receive for each option you and your partner agree on. You and your partner may settle upon any of the five options for each of the issues. Thus, there is a large number of possible settlements.

Again, remember that your goal is to earn as many points as possible, but that you **must** agree with your partner on each issue in order to earn points.

POINT STRUCTURE		
	Lowest	Highest
Robot's name	0	2400
Division of work responsibilities	−6000	0
Robot's size	−2400	0
Which competition to enter	0	1200
Robot's design	0	4000
Determination of robot parts	0	1600
Total	−8400	9200

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